Eye gaze as a window into good-enough processing: Comprehension of active transitive and suffixal passive constructions by Mandarin-Chinese-speaking learners of Korean

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Good-enough processing maintains that, given the parallel activation of two parsing routes—heuristics and algorithms, comprehenders often prioritise heuristics over algorithms, especially when input is noisy or resource demands are high (Christianson, 2016; Ferreira, 2003). Extending this framework to an under-explored L2 population and a typologically distinct language pairing, we investigate how L1-Mandarin-Chinese L2-Korean learners integrate word order, case-marking, and verbal morphology when processing active transitives and suffixal passives. Korean suffixal passives feature a NOM-marked theme, a DAT-marked agent, and passive verbal morphology (**PSV**; Table 1). **PSV** reverses initial associations between thematic roles and case markers (agent-NOM + recipient-DAT → theme-NOM + agent-DAT), pitting a THEME-FIRST strategy (1st NP = theme) against an AGENT-FIRST bias, and NOM-as-THEME / DAT-as-AGENT strategies against active-voice pairings (agent–NOM; theme–ACC). We ask how learners recruit morphology-driven algorithmic parsing under entrenched active-voice knowledge and weaker passive-voice strategies.

Methods. A webcam eye-tracking experiment (WebGazer.js; Papoutsaki et al., 2016) recorded eye movements of 49 L1-Mandarin-Chinese L2-Korean learners (**CHN**) and 56 native Korean speakers (**NSK**) during picture-matching of auditorily presented Korean sentences. We manipulated Voice (active transitive vs suffixal passive), Word Order (canonical vs scrambled), and Verb Position (verb-final vs verb-initial) in fully case-marked sentences. Participants selected the picture matching the sentence. Picture-selection responses were analysed via logistic mixed-effects modelling, fixation proportions to target vs competitor pictures in 50-ms bins using non-parametric permutation analysis (Abbot-Smith et al., 2017; Garcia et al., 2021), and **CHN**-only analyses on eye-gaze and proficiency were conducted via generalised additive mixed modelling (Lin & Zhang, 1999) within each region of interest (NP1, NP2, VP).

Results: Picture selection (Figure 1 & Table 2) **CHN** showed near-target performance only for canonical active transitives; accuracy dropped sharply for scrambled actives and was below chance across all suffixal passives. Verb position did not affect learners' performance, suggesting a limited role for verbal morphology in their interpretation. Proficiency selectively improved performance in structurally demanding conditions (scrambled actives; most passive conditions).

Results: Eye-tracking (Figure 2 & Table 3) **CHN** showed reduced target looks relative to **NSK** at critical regions, alongside garden-path patterns (e.g., the middle of NP1 in psv_cncl_vf), consistent with their reliance on the AGENT-FIRST bias and the agent–NOM mapping. Fronted **PSV** in verb-initial passives facilitated early theme-first interpretations for **NSK** but exerted only weak, transient effects for **CHN**. Proficiency modulated these dynamics in a region- and construction-specific manner—most strongly in (esp. scrambled) passives and less so in maximally variable word orders. This suggests that higher proficiency may support more efficient reanalysis and thematic-role integration, yielding transient success in exploiting **PSV**, but with continued vulnerability to active-voice knowledge and substantial individual- and item-level variability.

Together, our results suggest that L2 comprehenders engage in good-enough processing, but their heuristic preferences are strongly anchored in L1-based expectations and entrenched L2 active-voice knowledge. Case-marking, a highly reliable cue for native speakers, often misleads learners by reinforcing the agent–NOM mapping that conflicts with infrequent suffixal passives. Early **PSV** in verb-initial frames benefits native speakers more than learners, whose algorithmic use of **PSV** remained fragile and proficiency-dependent. Learners' heuristic-dominant strategies appear to be gradually recalibrated in proportion to proficiency (as an L2-usage-experience proxy), potentially yielding more robust algorithmic parsing. These findings extend good-enough processing accounts to a typologically distant L1–L2 pairing, highlight how cue reliability and cross-linguistic congruency shape L2 parsing, and demonstrate the viability of webcam eye-tracking for investigating morphosyntactic processing in diverse learner populations.

Table 1. Active transitive and suffixal passive constructions in Korean

Type	Active transitive, canonical	Suffixal passive, canonical
Example	Ciwu-ka Mia-lul cap-ess-ta. Ciwu-NOM Mia-ACC catch-PST-DCL 'Ciwu caught Mia.'	Ciwu-ka Mia-hanthey cap-hi-ess-ta. Ciwu-NOM Mia-DAT catch-**PSV**-PST-DCL 'Ciwu was caught by Mia.'
Thematic role ordering	*agent–theme* (scrambled: *theme–agent*)	*theme–agent* (scrambled: *agent–theme*)
Case-marking	Typical (agent–NOM; theme–ACC)	Atypical (theme–NOM; agent–DAT)
Verbal morphology	No	Yes (*-i/hi/li/ki-*)

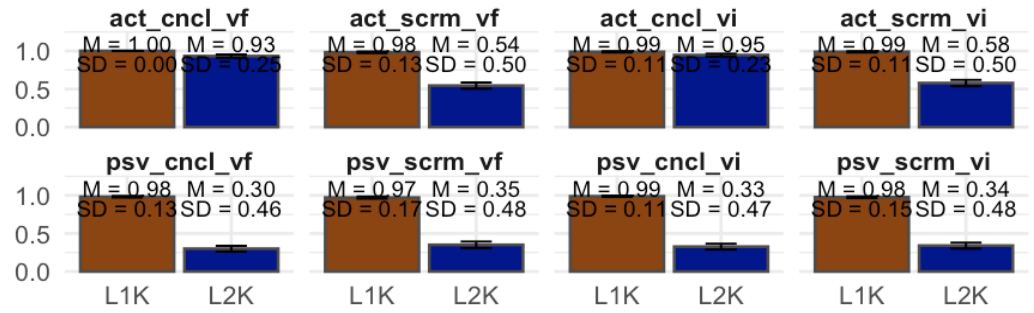

Figure 1. Results: Picture selection. **Brown = NSK**; **Blue = CHN**.

Table 2. Statistical outputs (picture selection): CHN only. $^{**}p<0.01$; $^{***}p<0.001$

	Verb-final				Verb-initial			
	act_cncl	act_scrm	psv_cncl	psv_scrm	act_cncl	act_scrm	psv_cncl	psv_scrm
Proficiency	*ns*	***	**	*ns*	*ns*	**	**	***

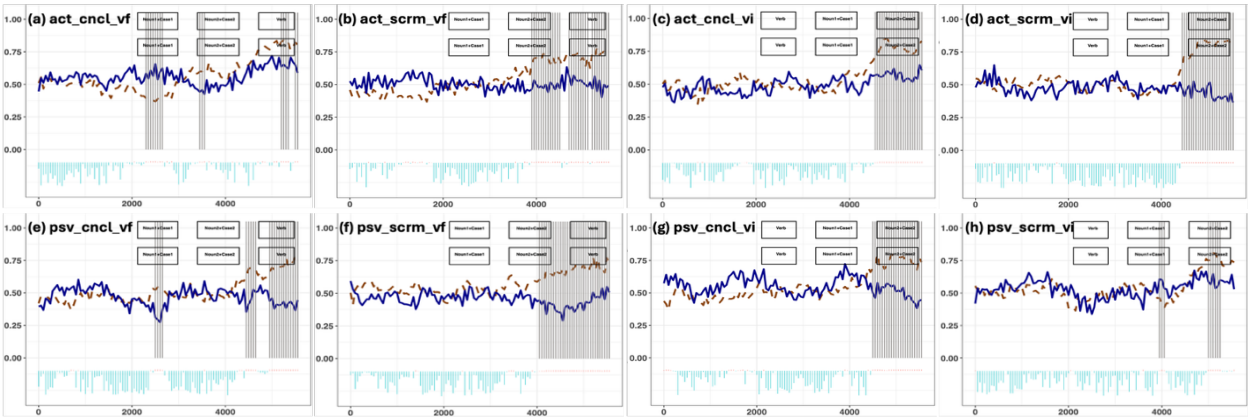

Figure 2. Results: Eye gaze. X-axis: time (in ms); Y-axis: proportion of looks to the target. **Brown = NSK**; **Blue = CHN**. Sentence regions are indicated by rectangles. Small bars below 0.00 represent linear regression *p*-values per 50-ms bin: light blue for $p > 0.05$ (non-significant), orange for $p < 0.05$ (significant). Grey shading above 0.00 denotes significant bins identified by permutation analysis.

Table 3. Statistical outputs (eye gaze): GAMM, CHN only. $^{*}p<0.05$; $^{***}p<0.001$

Smooth terms		Verb-final					Verb-initial			
		act_cncl	act_scrm	psv_cncl	psv_scrm		act_cncl	act_scrm	psv_cncl	psv_scrm
s(time): ctest	NP1	*ns*	*ns*	*ns*	***	VP	*ns*	*ns*	*ns*	*ns*
	NP2	*ns*	*	*	*ns*	NP1	*ns*	*ns*	*ns*	*ns*
	VP	***	*	*ns*	*	NP2	***	*ns*	***	*ns*

Note. *Time* was centred and *Ctest* was z-transformed for interpretability. Random effects included *Participant* and *Word*; all models showed significant s(participant) terms and most showed significant s(item) terms, indicating substantial by-participant/item variability within each region of interest.

Abbreviation: ACC = accusative case marker; DAT = dative marker; DCL = predicate ending (declarative); NOM = nominative case marker; PST = past tense marker; **PSV** = passive suffix.